ON A CURVE OVER AN ALGEBRAICALLY CLOSED FIELD TORSION COHOMOLOGY VANISHES IN DEGREES AT LEAST THREE, AND IN DEGREE TWO WHEN THE CURVE IS AFFINE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the separated case, the induction over an affine open covering, and the Mayer-Vietoris computation of the degree three group with the vanishing on the intersection.

Editorial note. The label theorem-vanishing-curves names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

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Allowed standard prerequisites (editorial).

- The precise vanishing theorem for separated curves. Same chapter, the vanishing and finiteness theorem for a separated finite type scheme of dimension one over an algebraically closed field. Supplies the separated case and the vanishing on the affine intersections.
- Mayer-Vietoris for a covering by two open subschemes. Same chapter, the Mayer-Vietoris exact sequence for an open cover. Supplies the exact sequence used to induct on the number of affine opens.

Source reference. Original reference 01EB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7176–7176. Prerequisite pointer invoked at proof line(s) 15756; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03SC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15656–15656. Prerequisite pointer invoked at proof line(s) 15752; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Theorem 0.1. *Let X be a finite type, dimension 1 scheme over an algebraically closed field k. Let $\mathcal{F}$ be a torsion sheaf on $X_{étale}$. Then*

$$H^q_{étale}(X, \mathcal{F}) = 0, \quad \forall q \geq 3.$$

If X affine then also $H^2_{étale}(X, \mathcal{F}) = 0$.

Proof. If X is separated, this follows immediately from the more precise Theorem 03SC. If X is nonseparated, choose an affine open covering $X = X_1 \cup \ldots \cup X_n$. By induction on n we may assume the vanishing holds over $U = X_1 \cup \ldots \cup X_{n-1}$. Then Mayer-Vietoris (Lemma 01EB) gives

$$H^2_{étale}(U, \mathcal{F}) \oplus H^2_{étale}(X_n, \mathcal{F}) \to H^2_{étale}(U \cap X_n, \mathcal{F}) \to H^3_{étale}(X, \mathcal{F}) \to 0$$

However, since $U \cap X_n$ is an open of an affine scheme and hence affine by our dimension assumption, the group $H^2_{étale}(U \cap X_n, \mathcal{F})$ vanishes by Theorem 03SC. □